

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-006 - Lakwa-006

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000064
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2003-12-16 / 2004-03-23
Measured / TVD	3560 m / 3548 m
Trajectory	Directional; inc 20.4 deg; az 333.6 deg
Formation Interval	1829-2879 m
Lithology	Shale-dominant interval with thin sandstone
Pore Pressure Reference	3742 psi
Fracture Gradient	15.22 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Critical (66.9/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	890	Surface
9 5/8 in casing	2064	Intermediate
7 in liner/casing	3132	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2004-02-10 2167 m	Stuck Pipe	Critical	Hookload became erratic during pipe movement. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2004-02-20 2500 m	Stuck Pipe	Critical	Hookload became erratic during pipe movement. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2004-01-02 775 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2004-02-27 2742 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2004-03-07 3034 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Check solids loading and BHA hydraulics before drilling ahead. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2003-12-31 697 m	Kick	Medium	Pit gain and flow with pumps off were observed. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2004-01-01 756 m	Pressure Spike	Medium	Pressure trend exceeded recent moving baseline. Mitigation: Check solids loading and BHA hydraulics before drilling ahead. Lesson: Correlate pressure anomalies with flow before escalating risk.

Completion Summary

The synthetic well reached 3560 m MD in the Kopili interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.